Agentic Retrieval Augmented Generation (RAG) Chatbot using large language models.

Abstract

Large Language Models (LLMs) are sophisticated artificial intelligence machines that are capable of comprehending and that can produce natural human text. These models that resemble deep learning architectures such as transformers are trained on large text corpora to automatically identify the patterns of language, its structure as well as its semantics. They are skilled in various natural language processing (NLP) subtasks such as text generation, summarization, translation, sentiment analysis, and question answering. Despite remarkable properties, Large Language Models (LLMs) face challenges that include providing false information, using obsolete facts and intricate reasoning mechanisms. Retrieval-Augmented Generation (RAG) seems to be the best approach through the integration of external knowledge databases. Such integration improves generation accuracy and credibility, particularly for tasks involving a lot of background knowledge and allows for continuous update and integration of domain specific facts. RAG combines LLM's innate understanding with access to enormous resource bases from external databases. This project aims to apply Retrieval-Augmented Generation to a specific domain, leveraging OpenAI's Generative Pretrained Transformer (GPT-3.5) model, and further customize it using LLM agents to develop a bespoke chatbot.